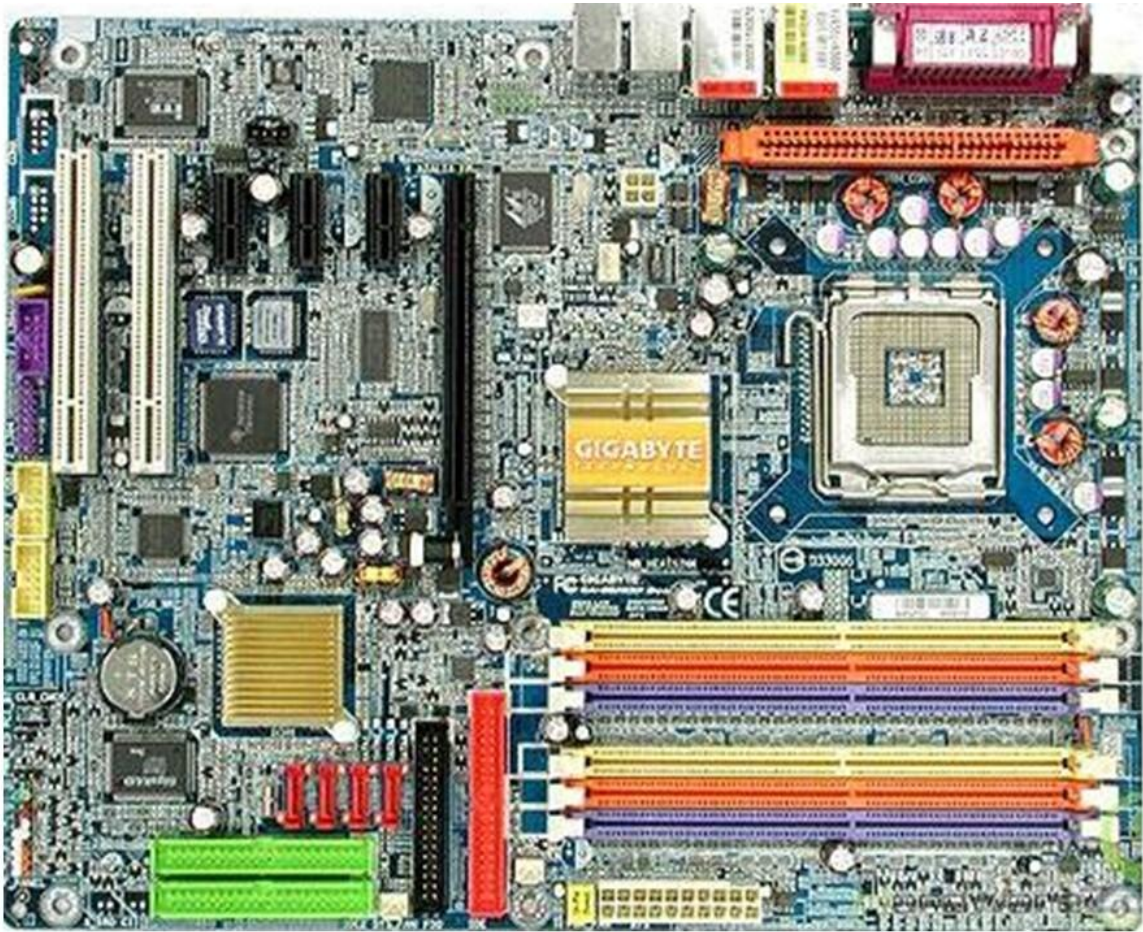
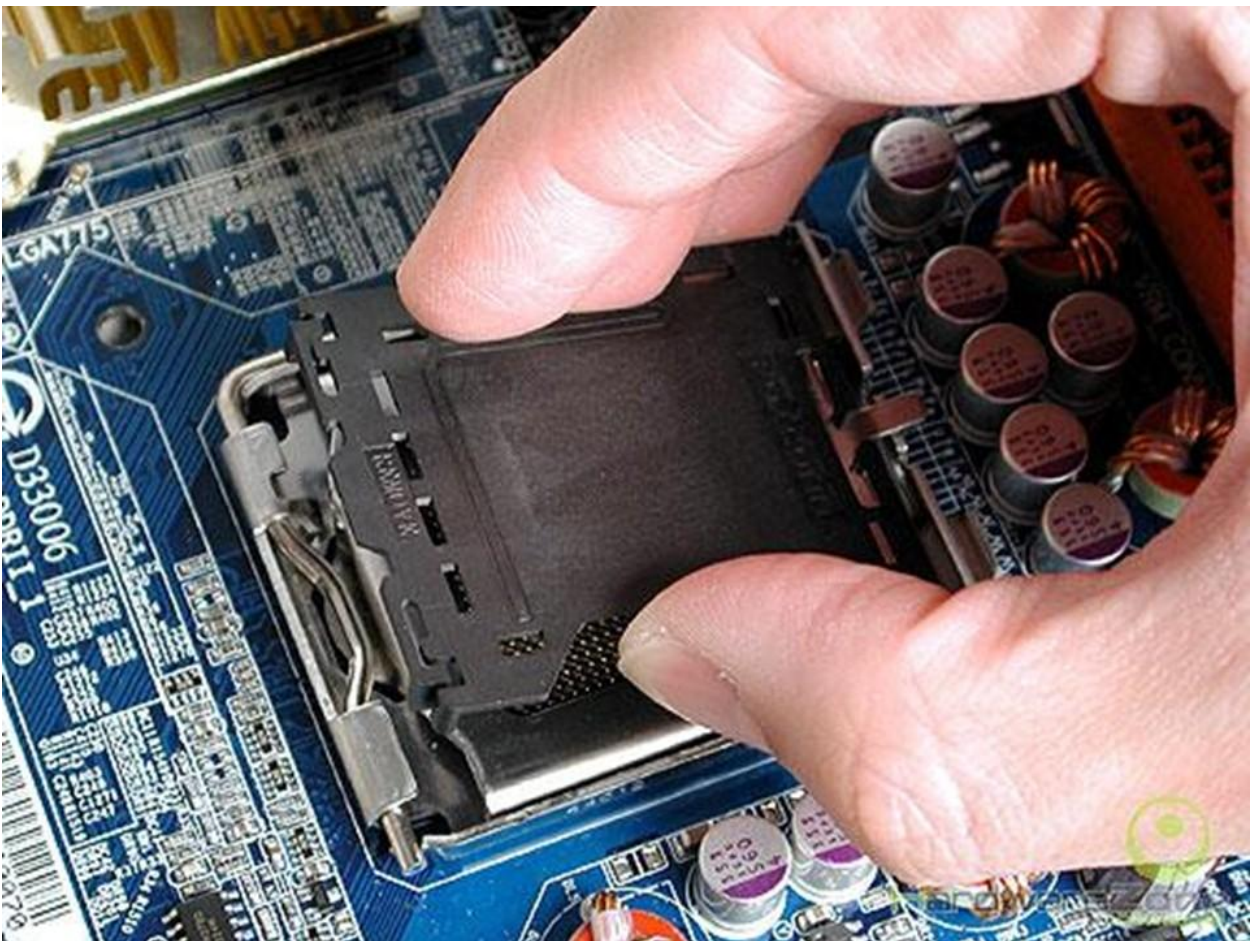
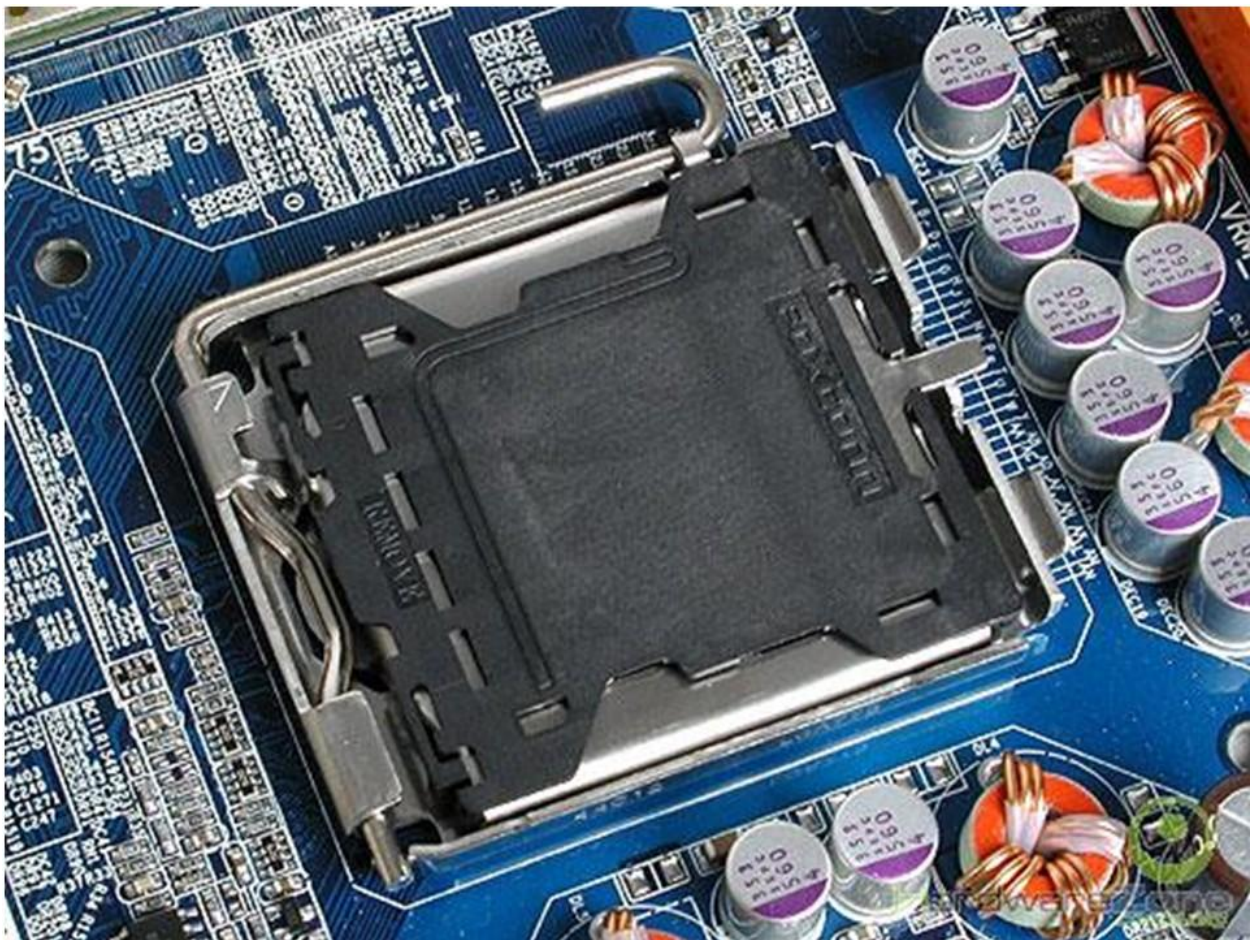


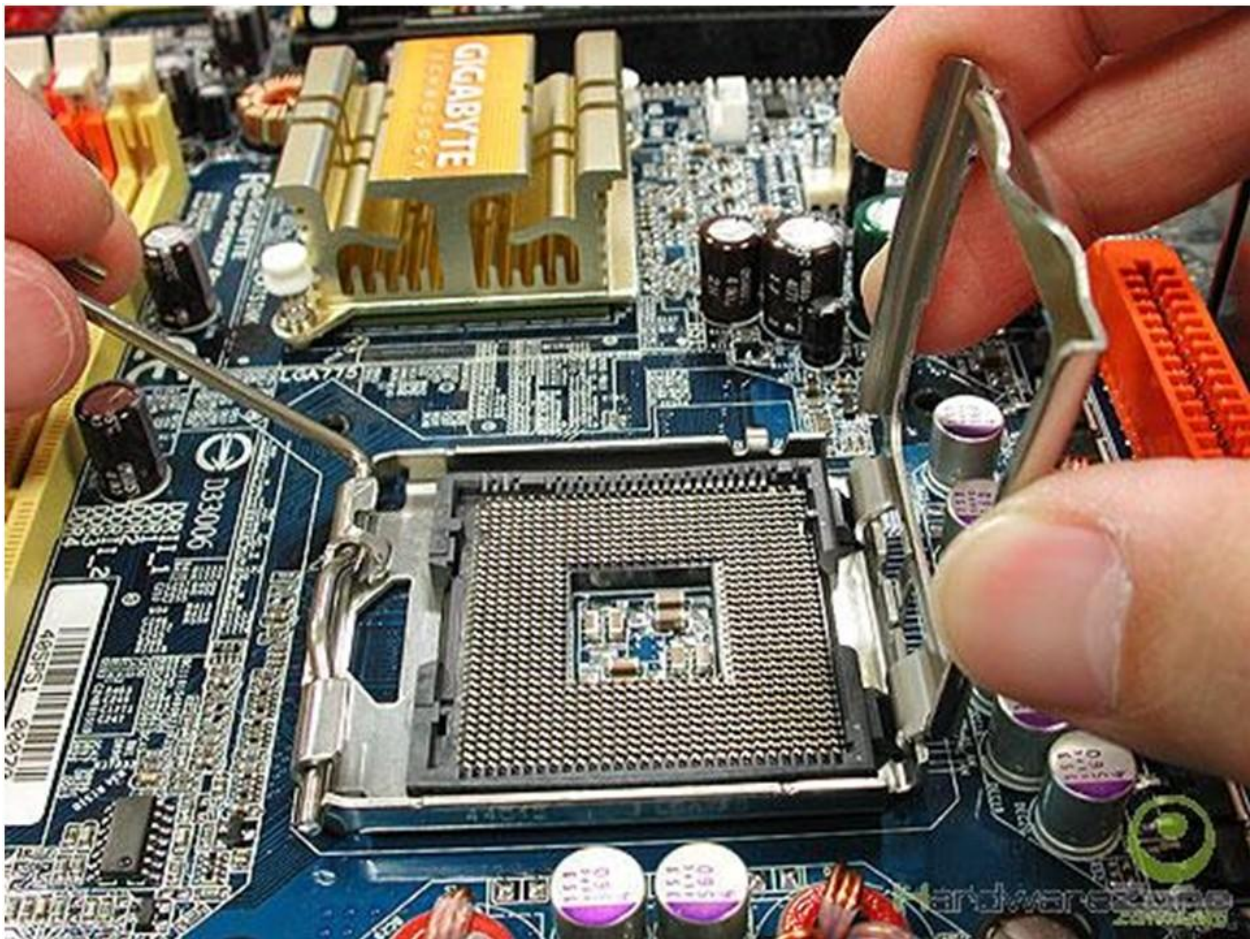
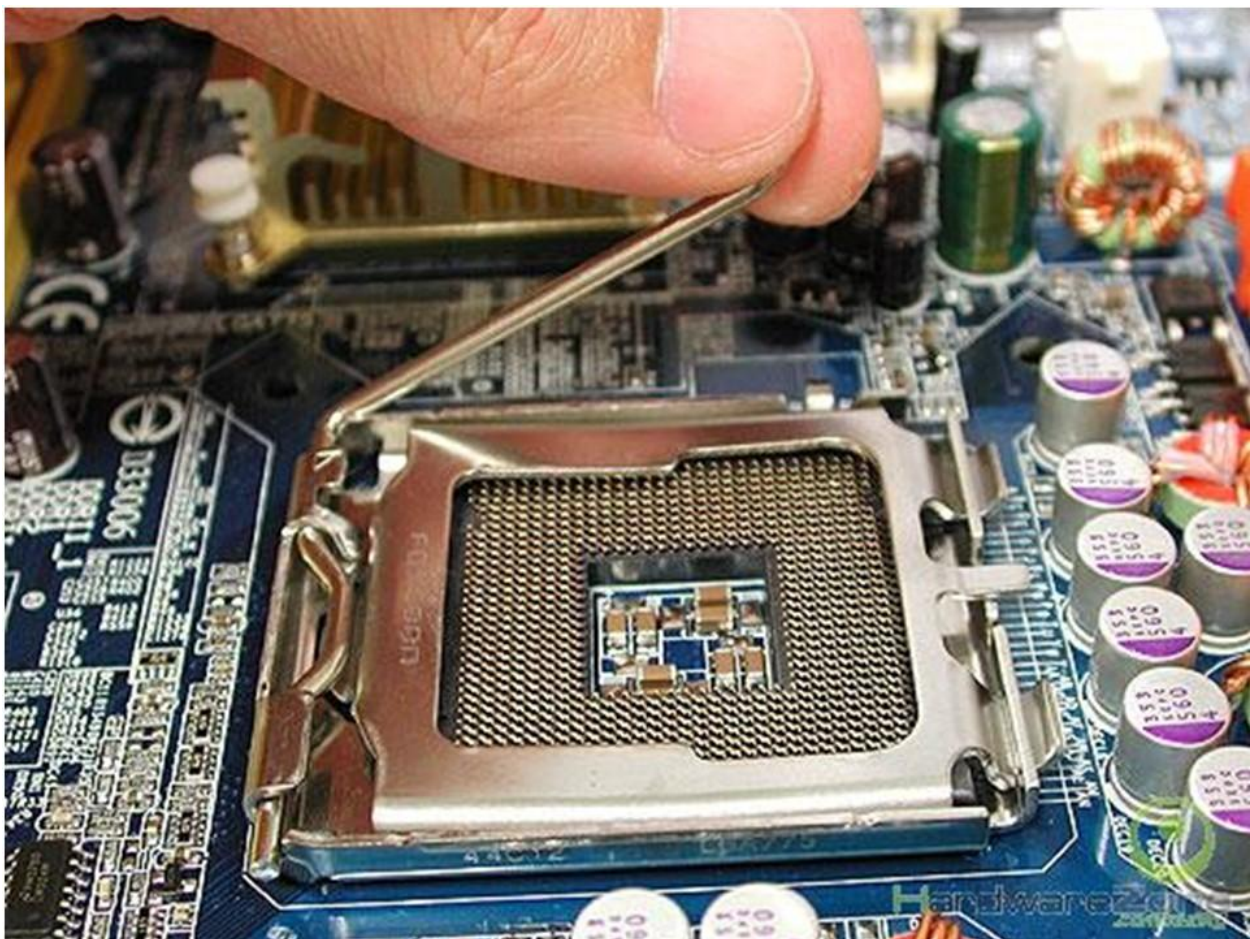
INSTALLING THE CPU

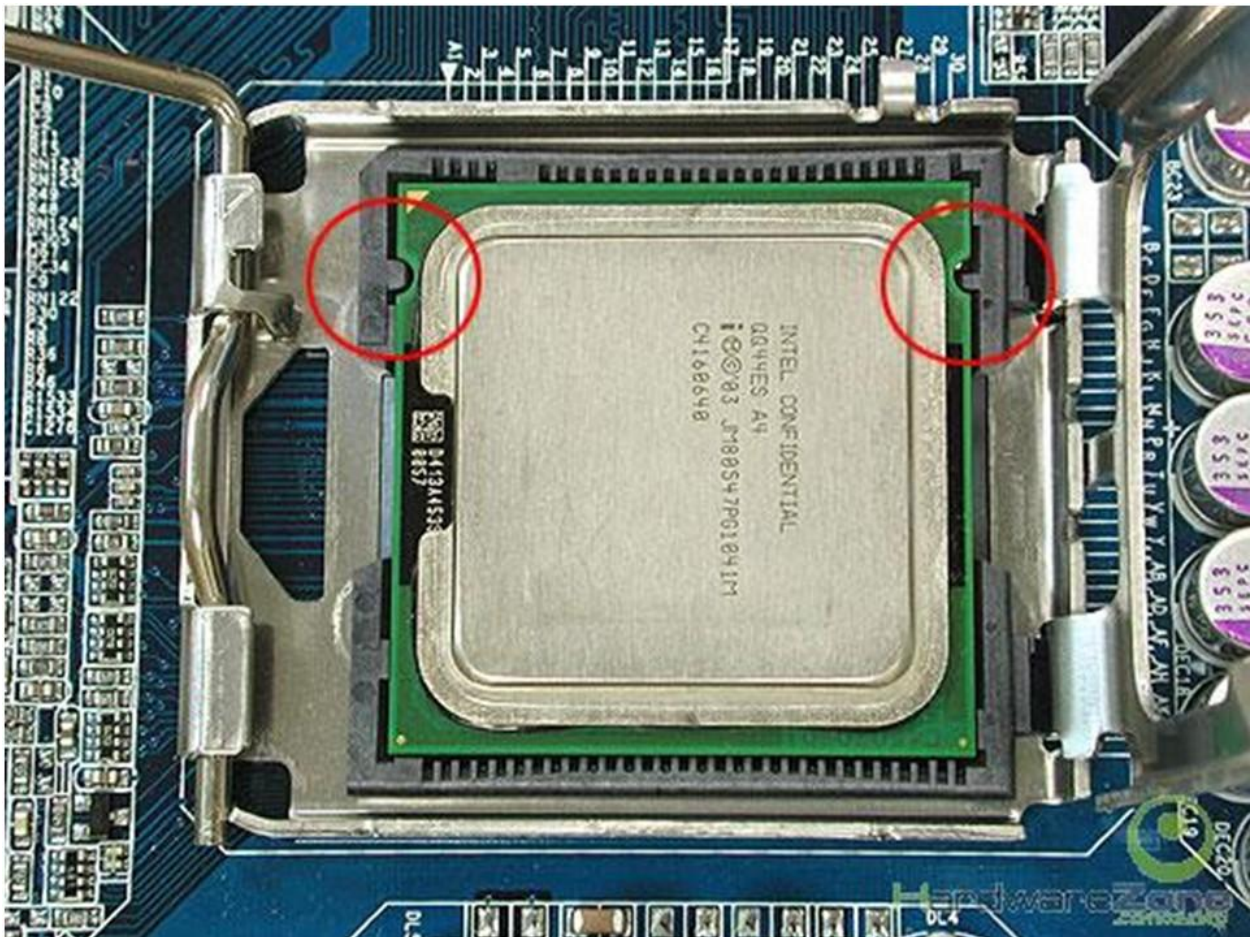
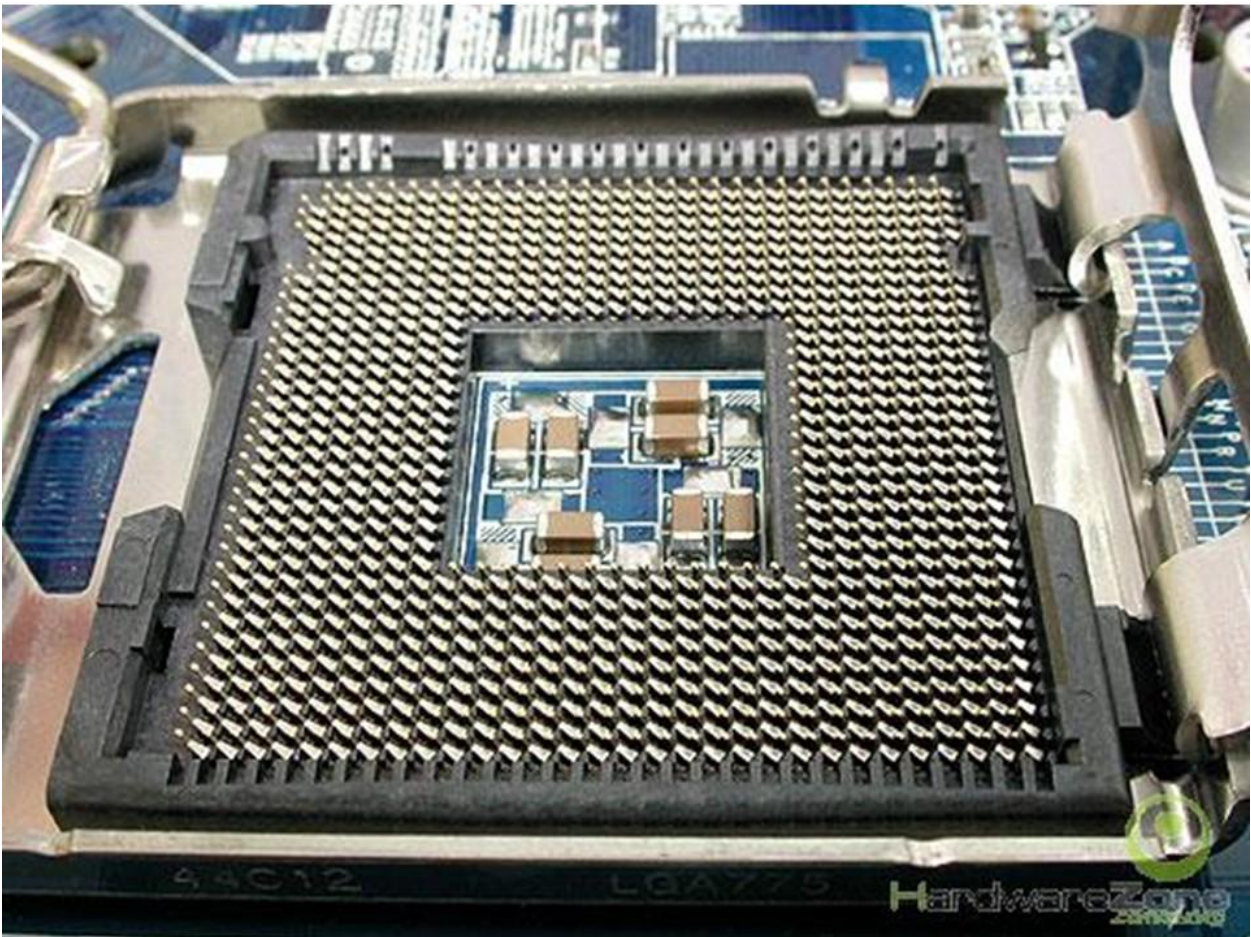
- INTEL -







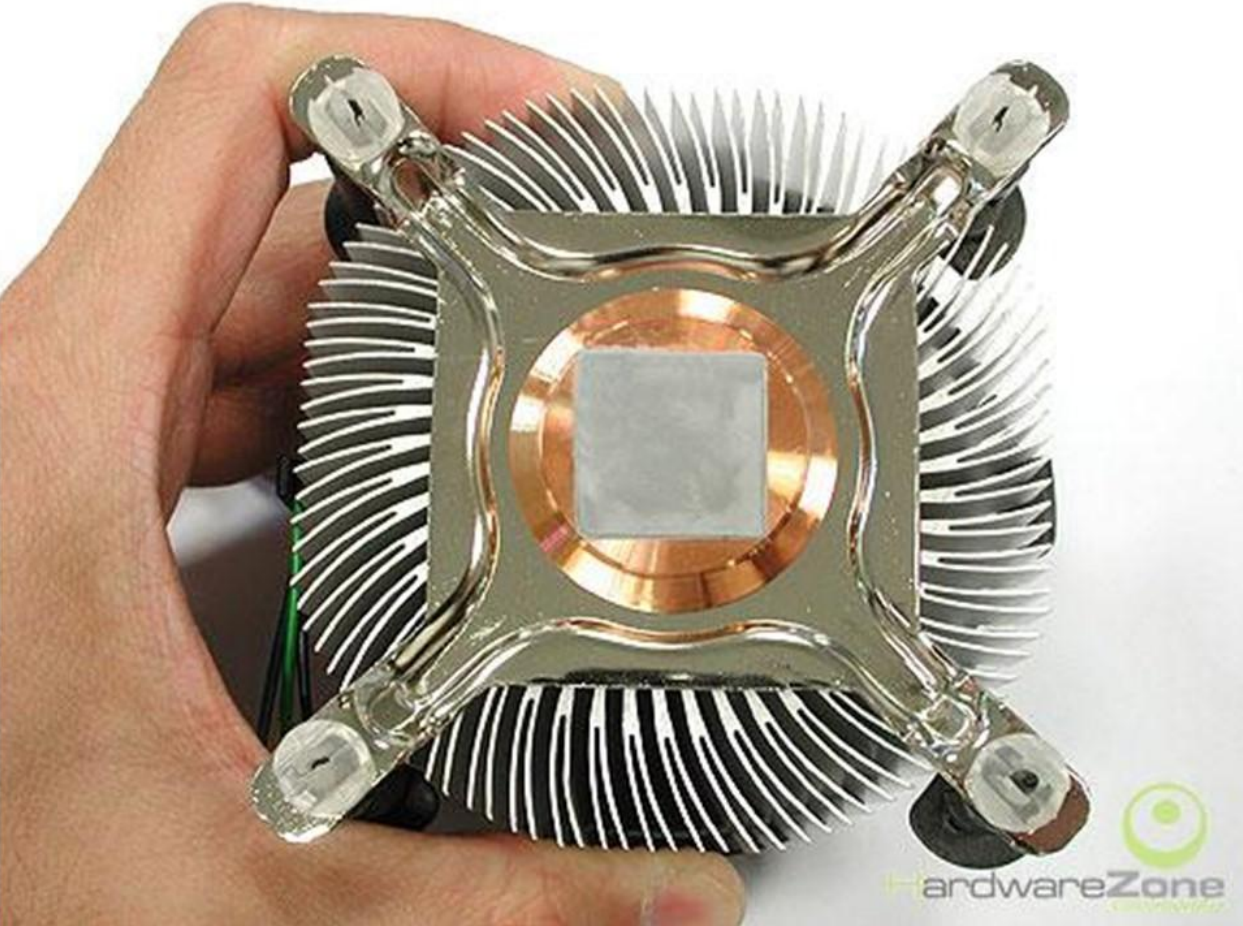


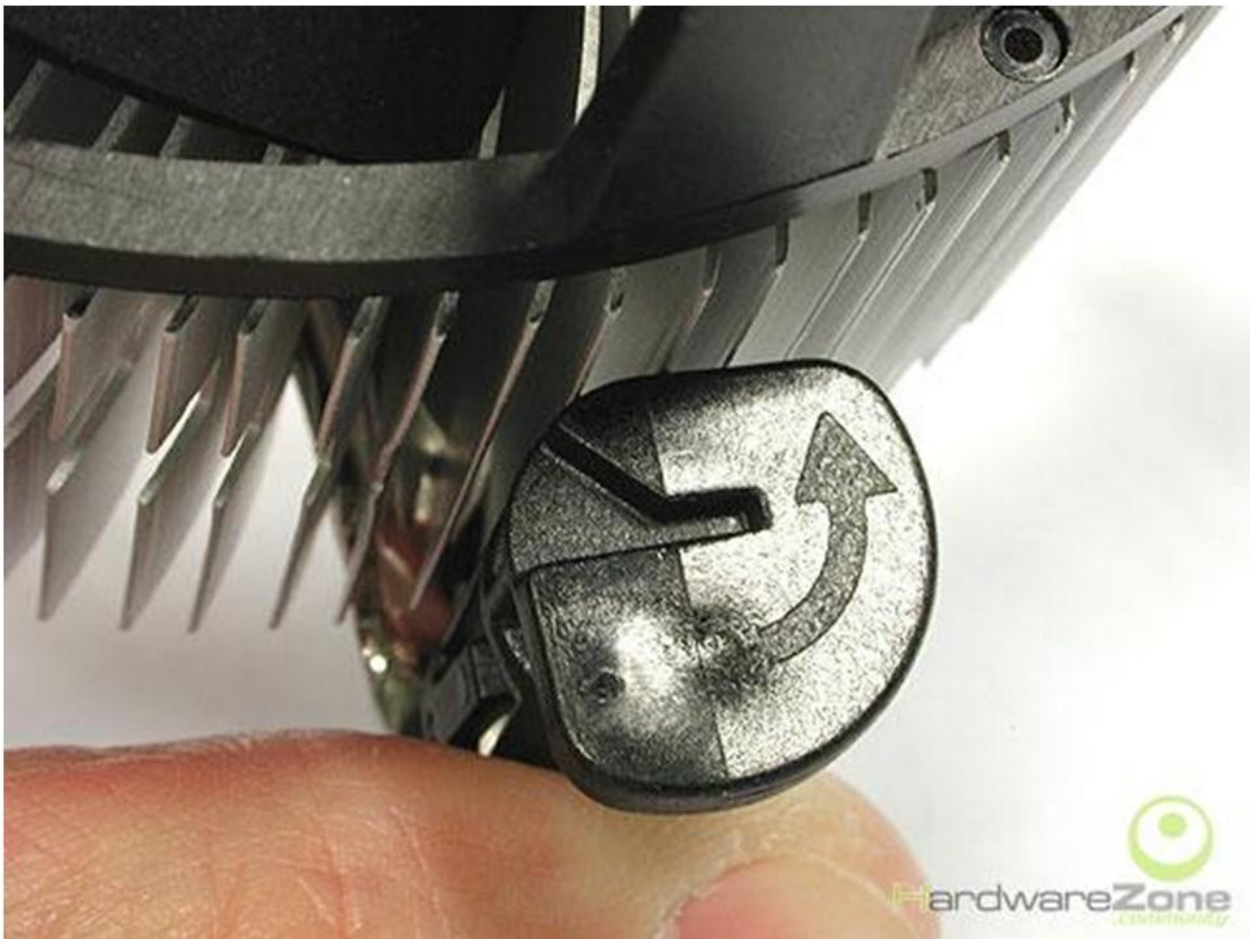


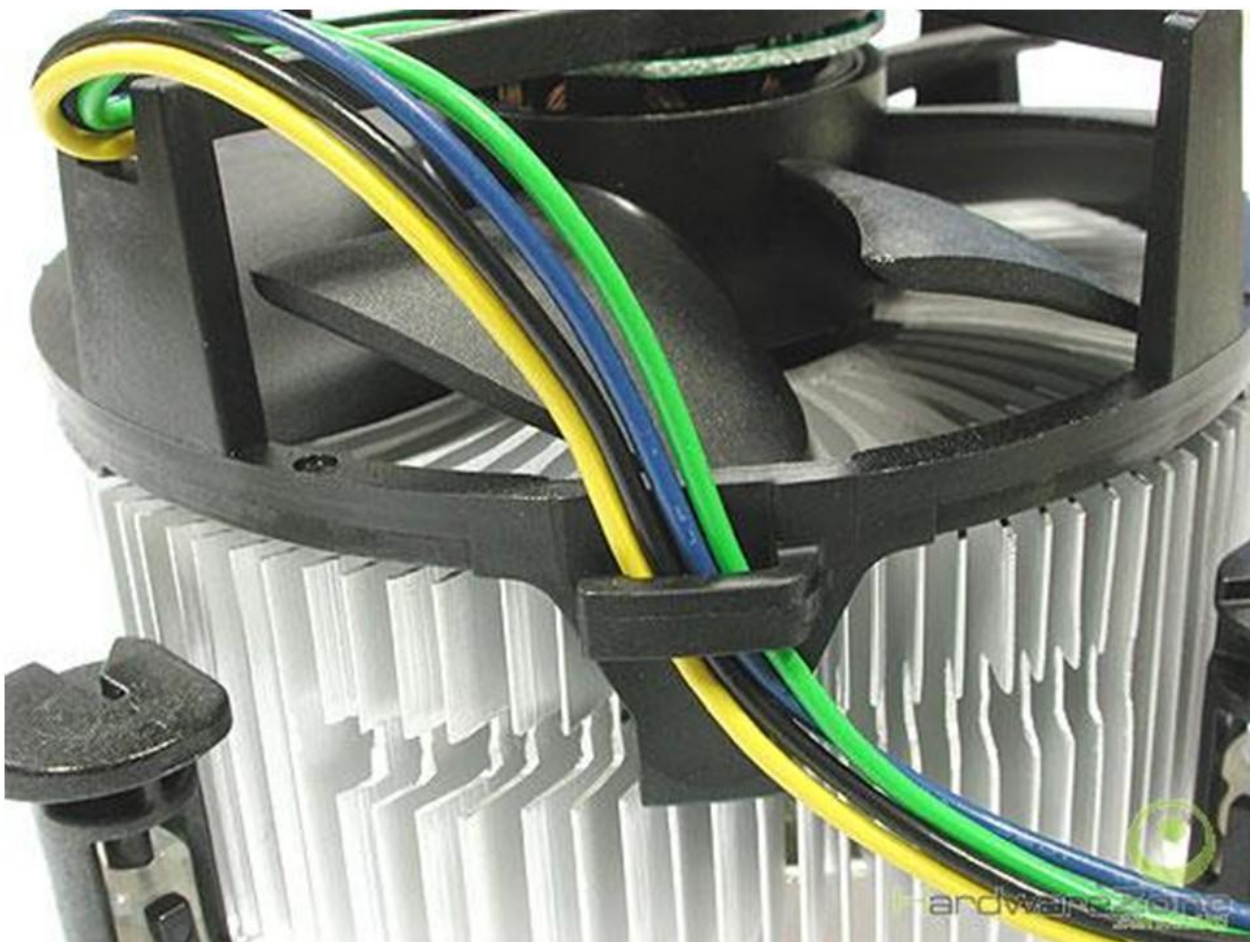


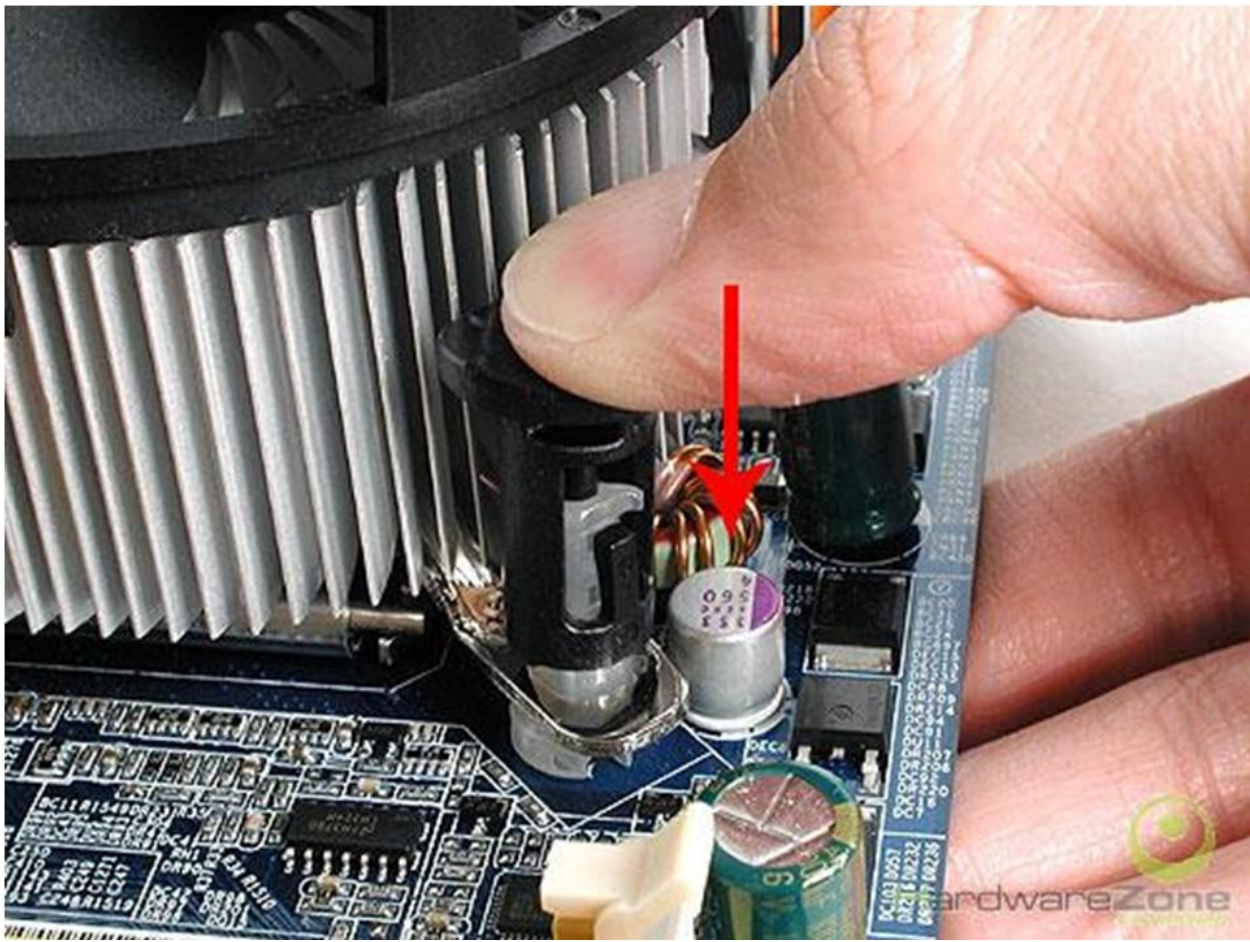
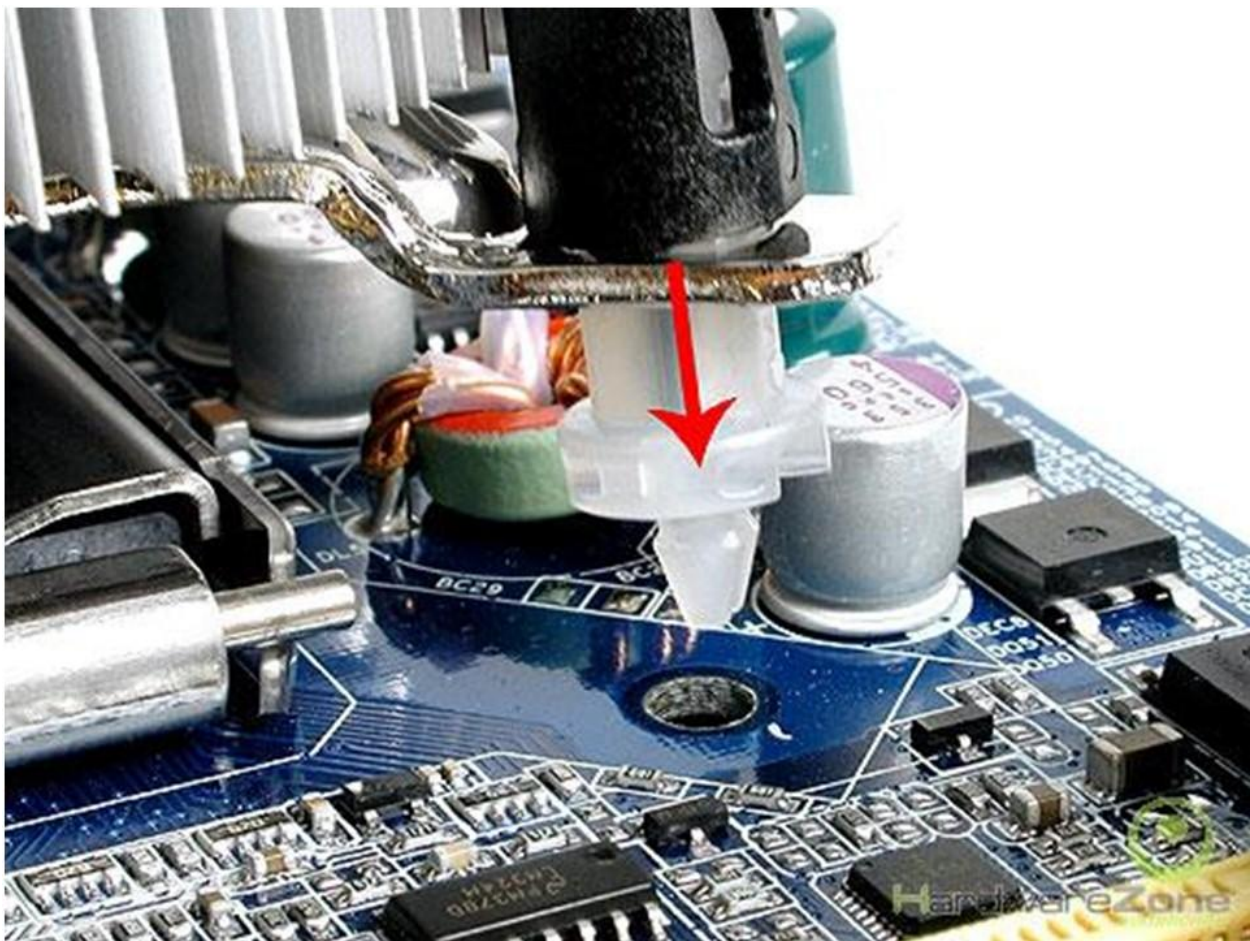


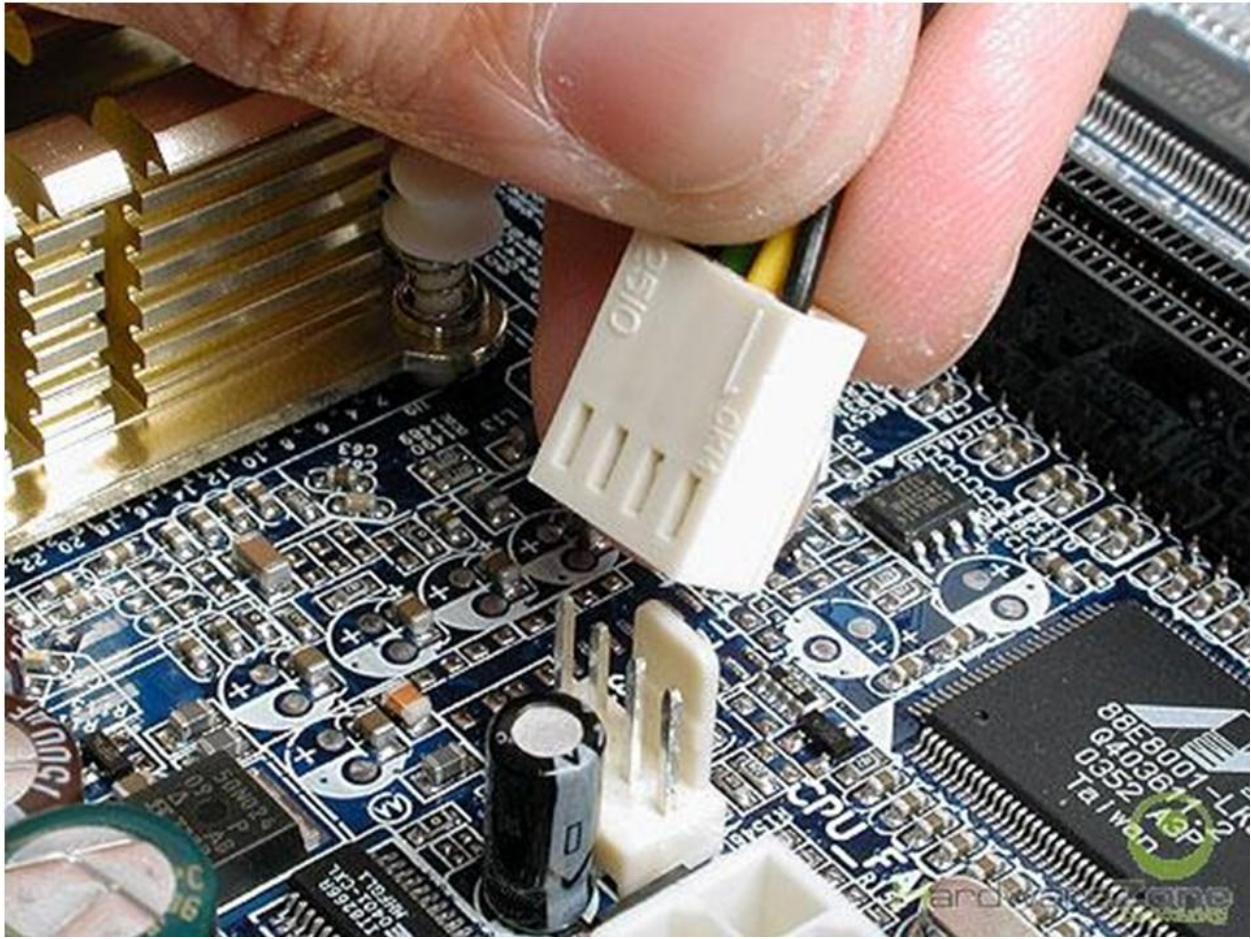
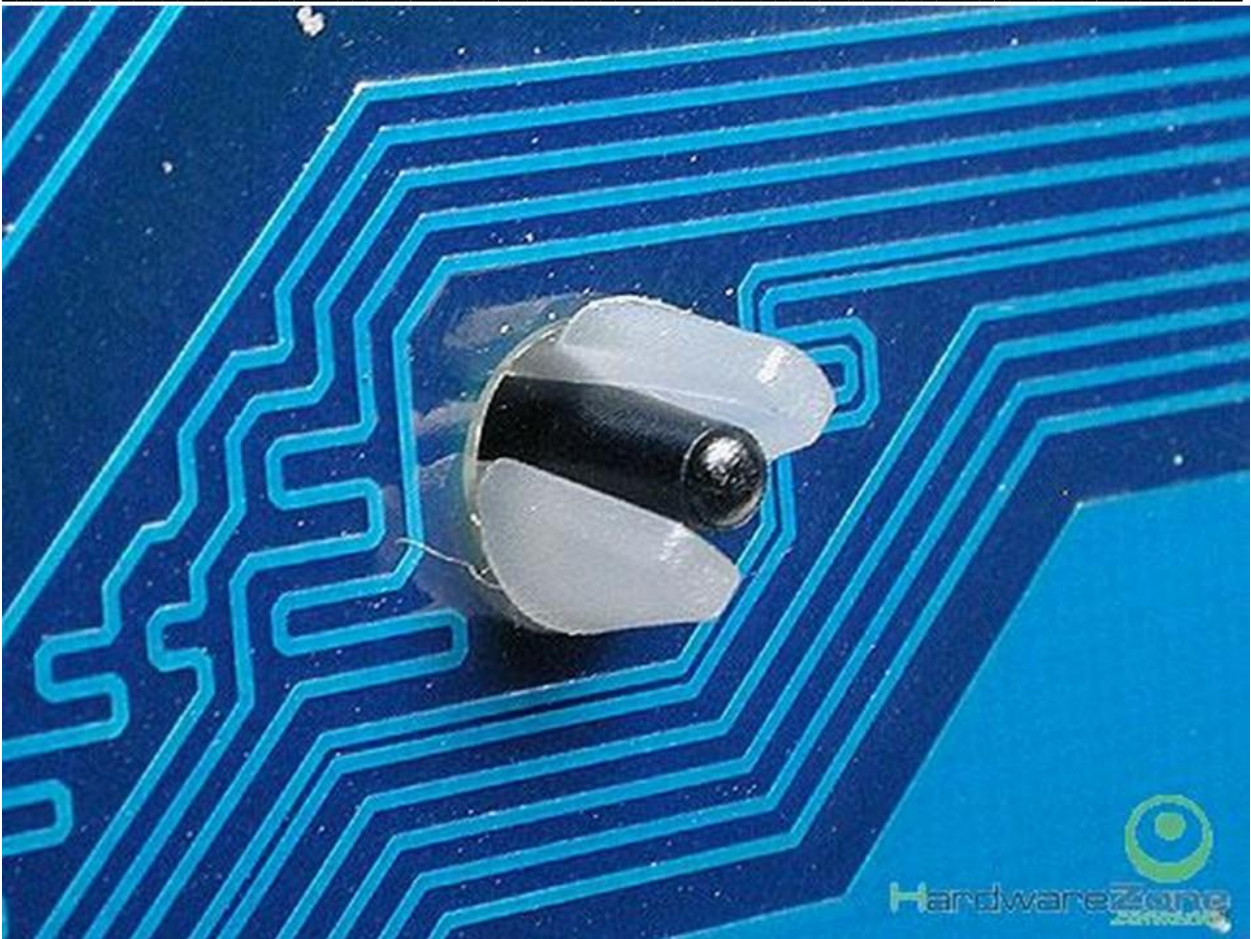
INSTALLING THE FAN / HEATSINK ASSEMBLY









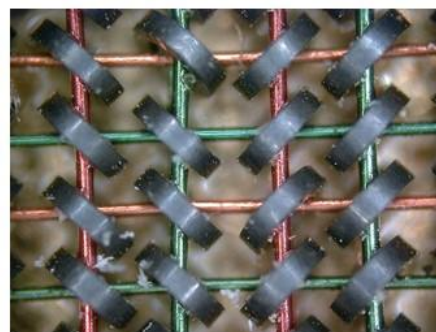
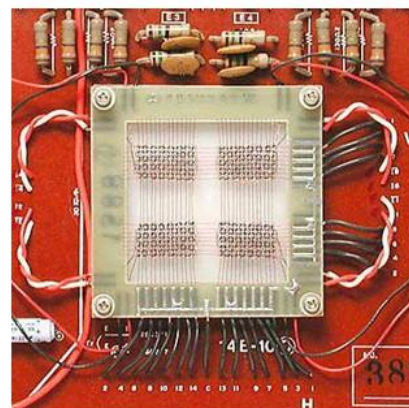




MEMORY

MEMORY

- commonly use the term "memory" to refer to RAM (Random Access Memory)
- The first type of random access memory was the magnetic core memory, developed in 1951
- A computer uses Ram to hold temporary instructions and data needed to complete tasks. This enables the computer's CPU (Central Processing Unit), to access instructions and data stored in memory very quickly.



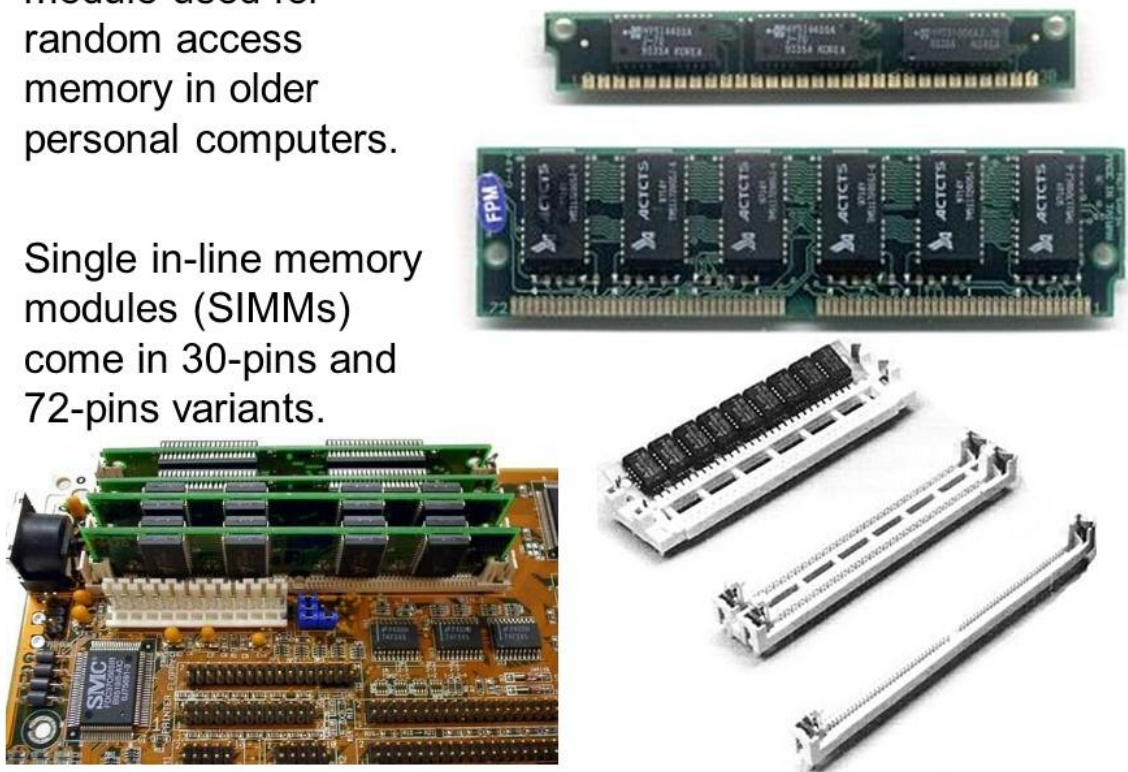
Two Types of Memory

- **RAM** - (Random Access Memory) is the area where your computer stores programs that you are currently running and data that you are currently working on.
- **ROM** - (Read Only Memory) is a class of storage media used in computers and other electronic devices. Because data stored in ROM cannot be modified

- is a type of memory module used for random access memory in older personal computers.

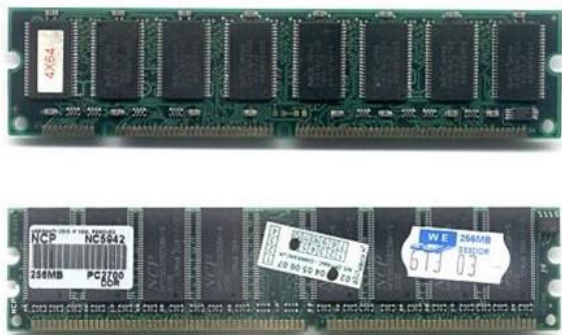
SIMM

- Single in-line memory modules (SIMMs) come in 30-pins and 72-pins variants.



DIMM

- Dual in line memory module
- This physical property describes a module with signal and power pins on both sides of the board.
- DIMMs can be any number of pins including 144pin SODIMM SDRAM, 168pin SDRAM memory, and 184pin DDR SDRAM.
- Transfer data 64 bits at a time
- Have no bottom-corner notch
- Have semicircular holes on both sides
- Have two notches among the contacts



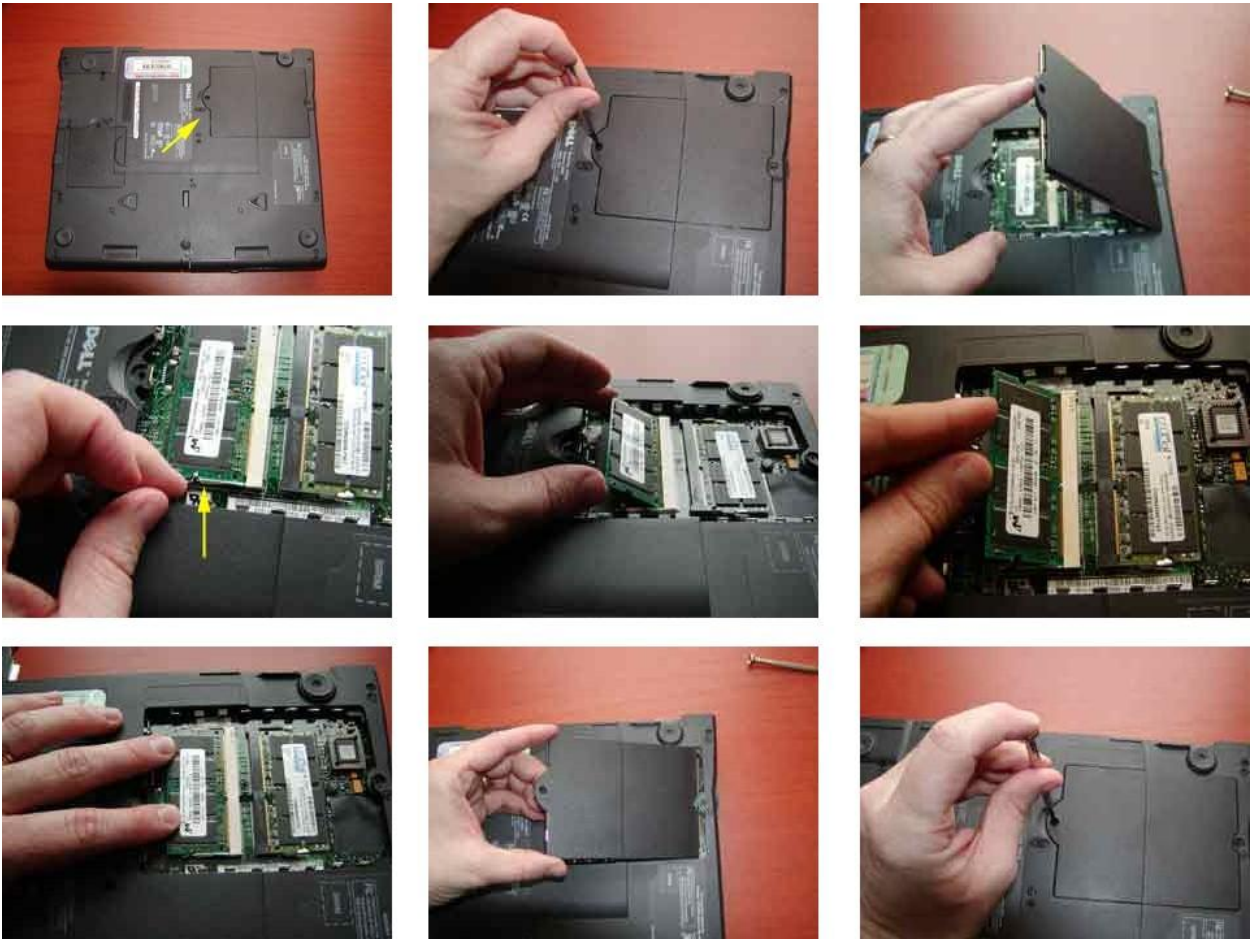
Two types of DIMMs: a 168-pin **SDRAM** module (top) and a 184-pin **DDR SDRAM** module (bottom).

SO-DIMM Chips

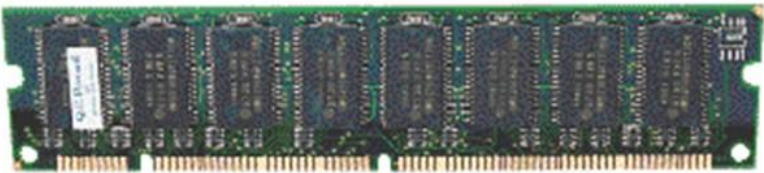
- Small outline dual in line memory module
- Are commonly used in notebook computers
- Smaller than standard DIMMs
- SO-DIMMs have 72, 100, 144, or 200 pins.
- 100-pin SO-DIMMs have two notches
- 144-pin SO-DIMMs have a single notch near (but not at) the center
- 200-pin SO-DIMMs have a single notch nearer to one side



How to Install SO-DIMM Chips?



SDRAM DIMM



- *Synchronous Dynamic Random Access Memory* Dual in-line memory modules have 168 pins
- SDR SDRAM, Originally simply known as "SDRAM"
- Voltage 3.3
- CAPACITY : 32MB , 64MB , 128 MB , 256 MB, 512

SDRAM Module :

PC – 66
PC – 100
PC – 133

Frequency Speed:

66 MHz
100 MHz
133 MHz

DRAM (RDRAM)

- Dynamic random access memory
- sometimes just called **Rambus DRAM** or **RDRAM**
- a 16-bit bus using a 184-pin RIMM
- they supported PC-800 RDRAM, which operated at 400 [MHz](#)
- provide systems with 16MB to 2GB of memory capacity at speeds of up to 1066MHz.



RIMM

- RIMM is not an acronym and is a trademark of [Rambus](#) incorporated for the Direct Rambus or DRAM modules
- developed by Kingston Technology Company in conjunction with Intel and Rambus, Inc., RIMM is the trademarked name for a Direct Rambus in-line memory module.
- RIMMs are faster than DIMMs but generate more heat. As such, heat spreaders made of aluminum are required to prevent overheating.



Freq. Speed : 800 MHz

Capacity Available :

64 MB, 128 MB, 256 MB

SO-RIMM

- Small Outline
- is a 160-pin module
- is a type of memory commonly used in laptop computers



DDR DIMM

- **Double Data Rate Dual Inline Memory Module**
- This new memory technology increases performance by executing two actions per clock cycle
- has a 184-pin
- is single notched



• DDR RAM varies on size and speed:		
Capacity :	DDR Module:	Freq. Speed :
128 MB	PC – 2100	266 MHz
256 MB	PC – 2700	333 MHz
512 MB	PC – 3200	400 MHz
1 GB		

DDR SDRAM

- double data rate synchronous dynamic random access memory
- data being transferred 64 bits at a time
- the older version is sometimes referred to as DDR1-SDRAM
- has a 184-pin
- is single notched
- Voltage 2.4 or 2.5



184pin

- DDR RAM varies on size and speed:

Capacity :	Module name:	Freq. Speed :
128 MB	PC – 1600	200 MHz
256 MB	PC – 2100	266 MHz
512 MB	PC – 2400	300 MHz
1 G	PC – 2700	333 MHz
	PC – 3200	400 Mhz

DDR2 SDRAM

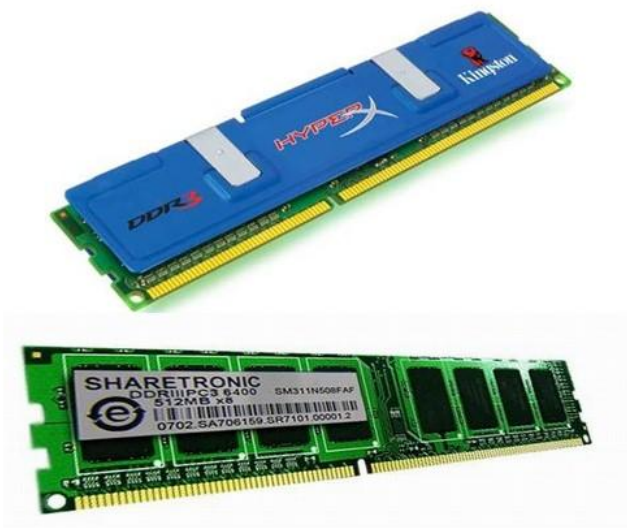


Capacity :	Module name:	Freq. Speed :
128 MB	PC – 3200	400 MHz
256 MB	PC – 4200/4300	533 MHz
512 MB	PC – 5300/5400	667 MHz
1 & 2 G	PC – 6400	800 MHz
	PC – 8500/8600	1066 MHz

- double-data-rate two synchronous dynamic random access memory
- is the ability to operate the external data bus twice as fast as DDR SDRAM
- is a 240-pin module (DDR2 SODIMMs are 200-pin)

DDR3 SDRAM

- double-data-rate three synchronous dynamic random access memory
- DDR3 SDRAM is an improvement over its predecessor, DDR2 SDRAM
- have 240 pins, the same number as DDR2, and are the same size, but are electrically incompatible and have a different key notch location



Capacity :

128 MB
256 MB
512 MB
1 & 2 G

Module name:

PC3 – 6400
PC3 – 8500
PC3 – 10600
PC3 – 12800

Freq. Speed :

800 MHz
1066 MHz
1333 MHz
1600 MHz



DDR



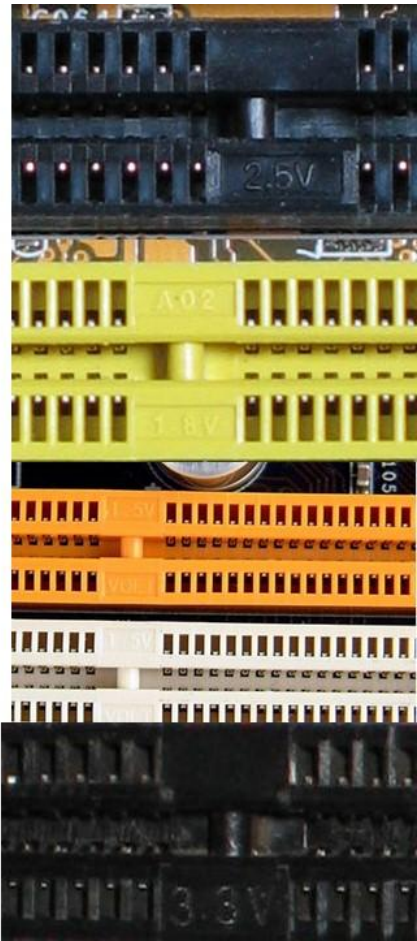
DDR 2



DDR3



SDRAM



- **Causes of Memory Errors**

- - mixing chips with different speed rating in one bank
- - mixing chips with different manufacturers in the same bank
- - poor chip connection
- - insufficient power and/or noisy power line
- - static electricity
- - improper shielded sources of RF noise

- **Installing Memory Chips**

- - beware of static electricity, if ground strap is not available; take your shoes and socks off.
- - do not remove or insert memory chips while the power is on
- - use a chip remover when removing chip
- - check chip seating

VIDEO DISPLAY ADAPTERS

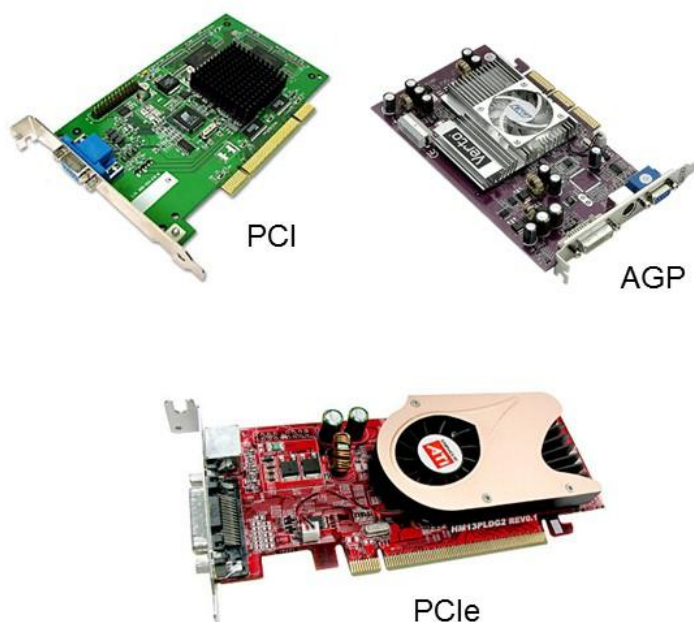
VIDEO CARD

- referred to as a graphics accelerator card, display adapter, graphics card
- Its function is to generate and output images to a display.
- is an expansion card that allows the computer to send graphical information to a video display device such as a monitor or projector.



Video Card/ Video Adapter / Display Card / Graphics Card

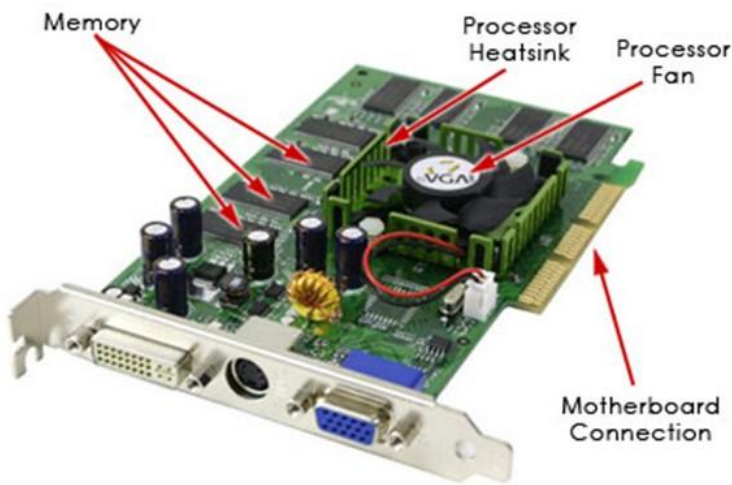
- Responsible for creating the picture displayed by a monitor
- Sometimes built in the motherboard
- Major manufacturer are ATI, NVidia, Asus
- ISA, PCI, AGP, or PCI Express



Video Card Package



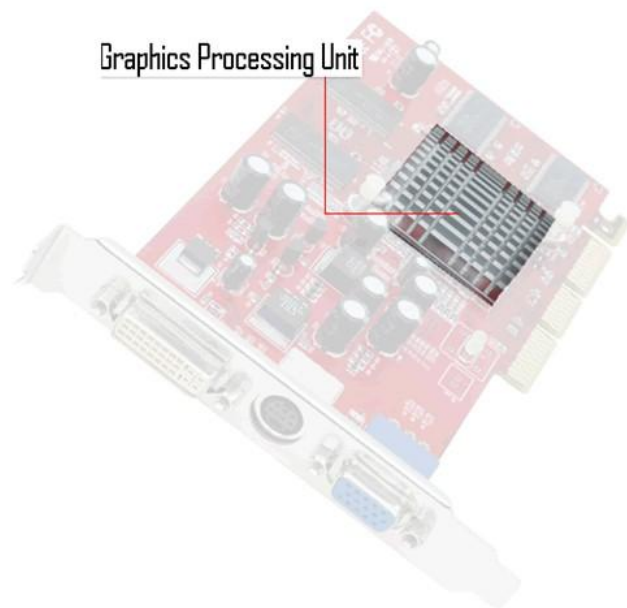
Four main components of a graphics card



The graphics card decides how to use the pixels on the screen to create the image. It then sends that information to the [monitor](#) through a cable.

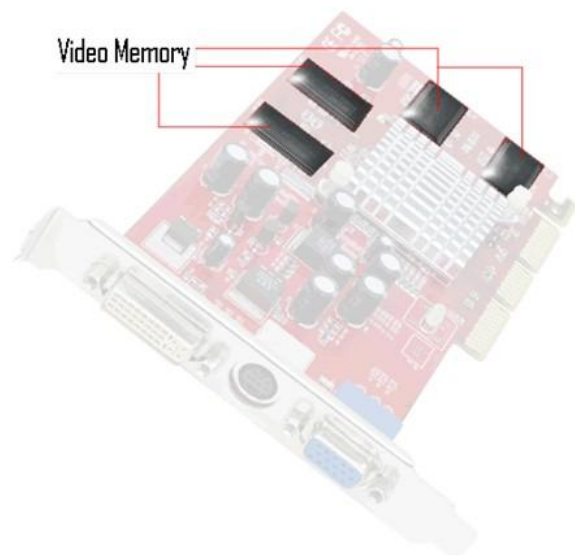
GPU

- **graphics processing unit**
- also called **visual processing unit** or **VPU**
- is a dedicated graphics rendering device for a personal computer, workstation, or game console.
- A GPU can sit on top of a video card, or it can be integrated directly into the motherboard.
- Executes the Instruction interpreted by the driver software



Video Memory

- The main use of the video memory is as the frame buffer. This is the place where the information is stored about the video image itself.
- used for storing other data as well as the screen image which manages the depth coordinates in [3D graphics](#).
- holds information about each pixel and to temporarily store completed pictures



RAMDAC

- Random Access Memory Digital-to-Analog Converter.
- takes responsibility for turning the digital signals produced by the computer processor into an analog signal which can be understood by the computer display.
- With CRT displays, it is best to work over 75 Hz and never under 60 Hz, in order to minimise flicker.

SLI Scalable Link Interface

- is a brand name for a multi-GPU solution developed by Nvidia for linking two or more video cards together to produce a single output.
- The basic idea of SLI is to allow two or more graphics processing units (GPUs) to share the work load when rendering a 3D scene.
- Ideally, two identical graphics cards are installed in a motherboard that contains two PCI-Express x16 slots, set up in a master-slave configuration.
- Both cards are given the same part of the 3D scene to render, but effectively half of the work load is sent to the slave card through a connector called the SLI Bridge.
- When the slave card is done, it sends its output to the master card, which combines the two images to form one and then outputs the final render to the monitor.



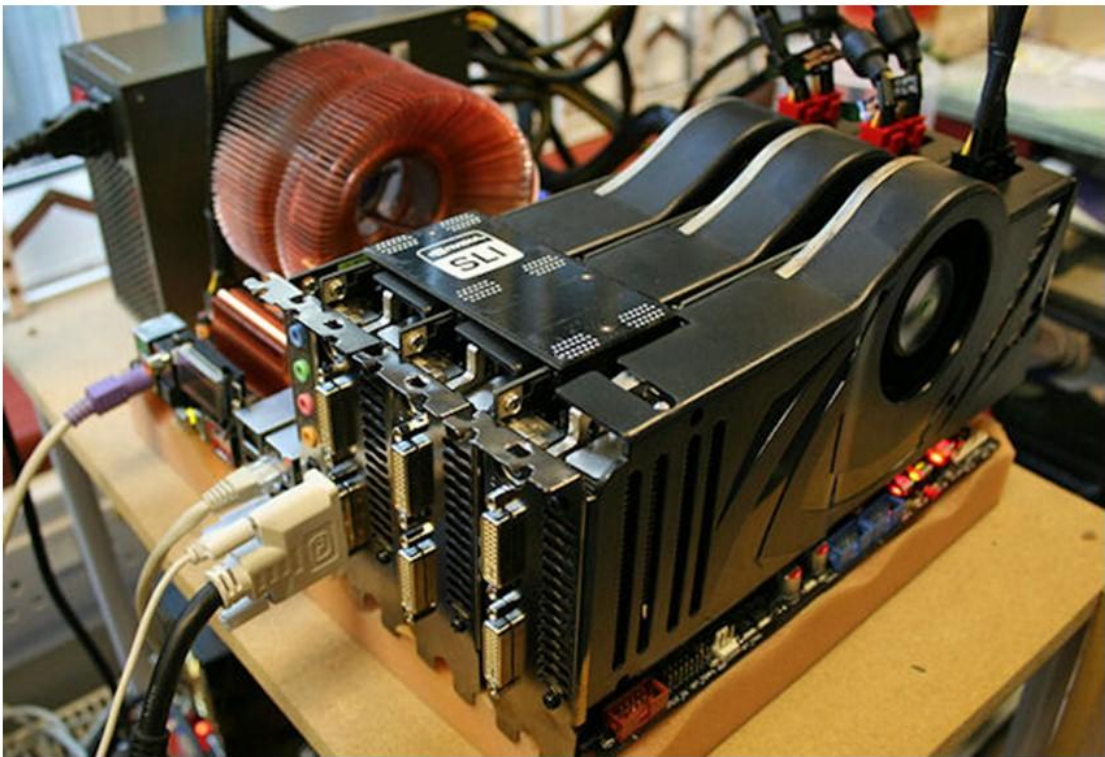
NVIDIA's SLI link card



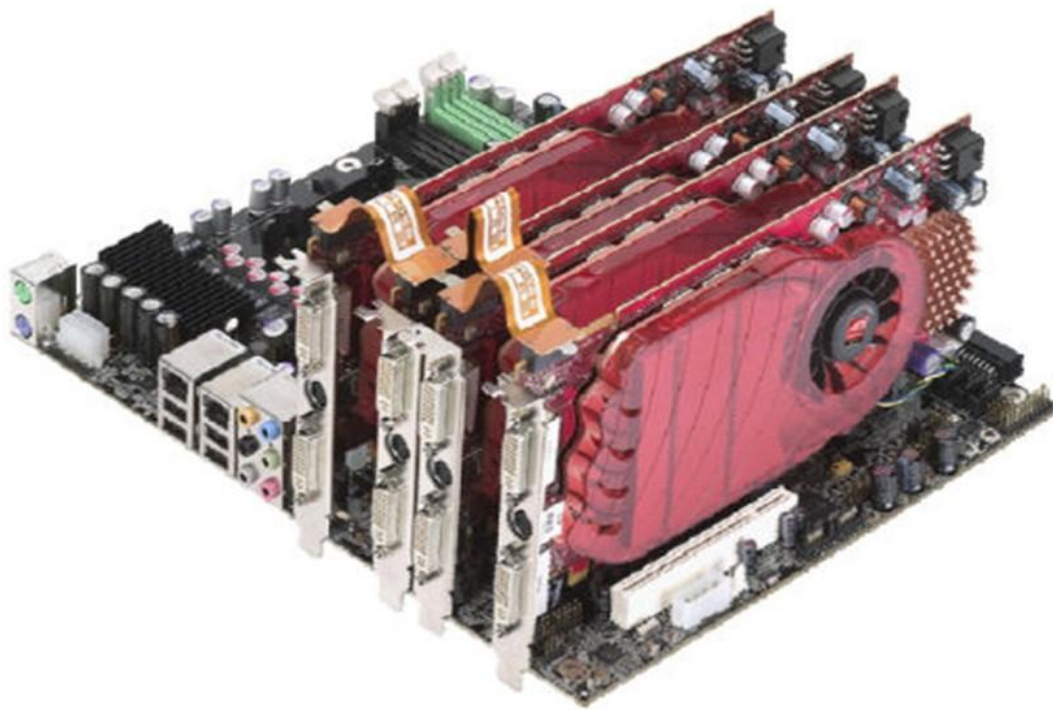
2 Way SLI



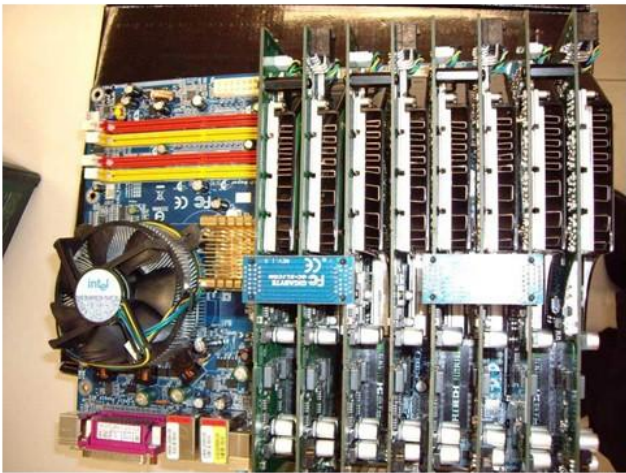
3 Way SLI



4 Way SLI



8 Way SLI



The most common connection systems between the video card and the computer display are:



- [HD-15](#): Analog-based standard adopted in the late 1980s designed for CRT displays, also called VGA
- [DVI](#): Digital-based standard designed for displays such as [LCDs](#), plasma screens and video projectors.
- [S-Video](#): Included to allow the connection with televisions, [DVD players](#), [video recorders](#) and [video game consoles](#).

Video Display



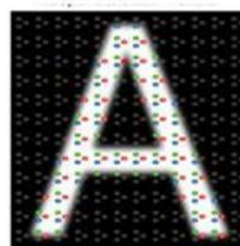
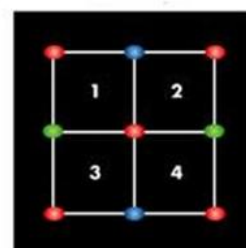
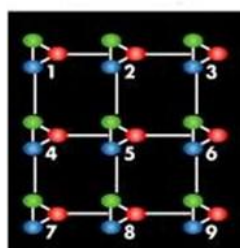
- The monitor itself should be compatible with the video adapter
- Primary device
- Uses either cathode ray tube (CRT) or Liquid Crystal Display (LCD)



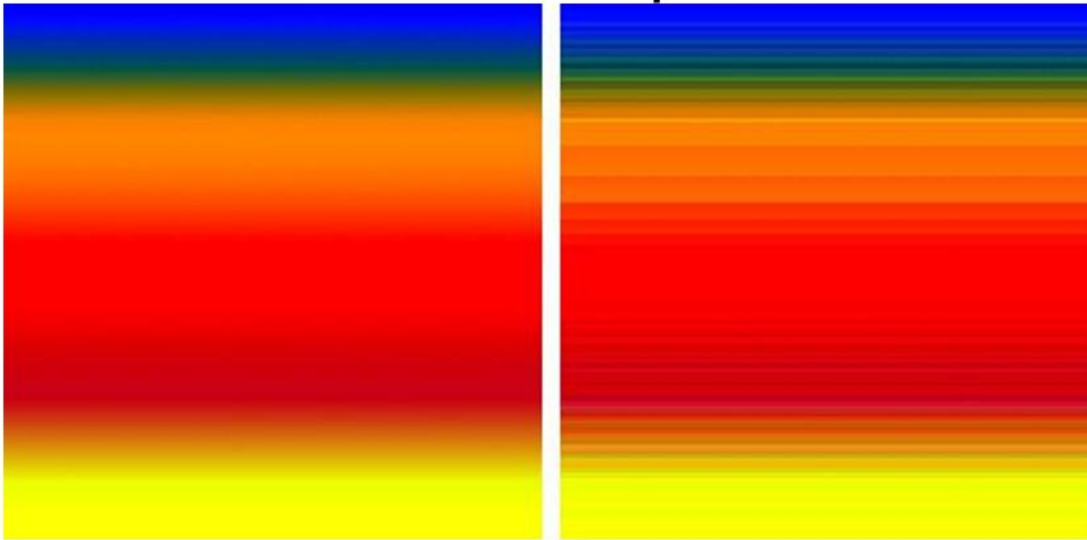
Video Terminology

Pixel

- are thousands of small dots composing a screen image
- It represent the smallest piece of the screen that can be controlled individually
- It can be set to a different color and intensity



Color Depth

**8 bit****16bit**

- The precise appearance of each pixel is controlled by the intensity of these beams of light
- The amount of information that is stored a pixel determines its color dept

Graphics

- A representation of image of drawing using the memory of the video card for each pixel (picture element) or dot on the screen



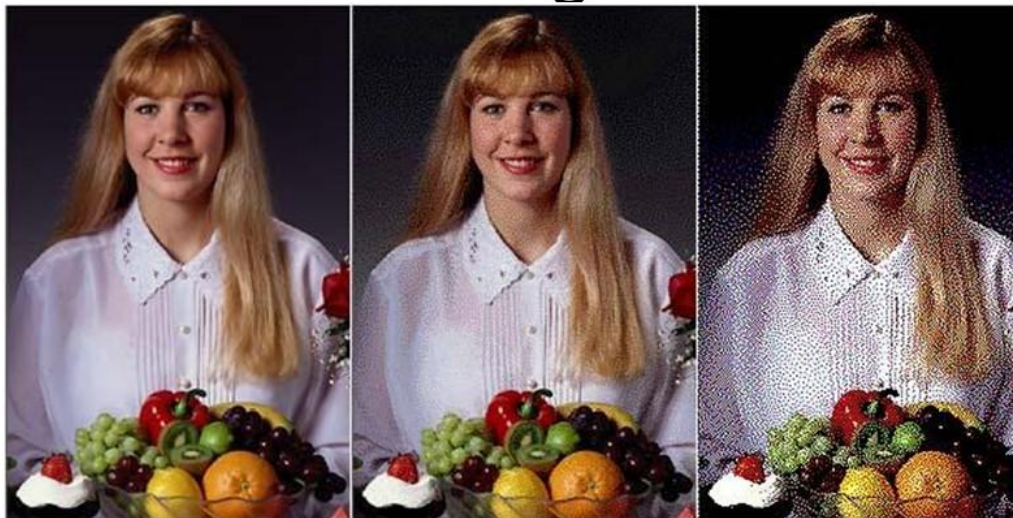
Resolution



- Refers to the quality of the screen
- The higher the resolution, the sharper the image on the screen
- The number of pixel displayed horizontally and vertically



Scanning Rate



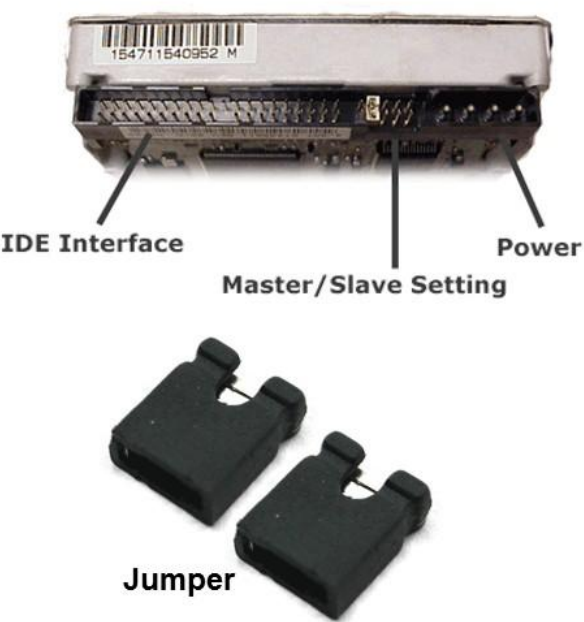
- The time it takes for the electron beam to draw a beam light
- The faster the scanning rate the higher the resolution

Text

- Predefined characters, numbers, special characters and extended characters
- Usually refereed to as the IBM characters extended ASCII characters
- American Standard Code for Information Interchange

0	32	sp	64	Q	96	ü	128	á	160	l	192	æ
1	33	!	65	A	97	û	129	â	161	l	193	æ
2	34	"	66	B	98	ü	130	á	162	l	194	æ
3	35	#	67	C	99	ü	131	â	163	l	195	æ
4	36	\$	68	D	100	ü	132	á	164	l	196	æ
5	37	%	69	E	101	ü	133	â	165	l	197	æ
6	38	&	70	F	102	ü	134	á	166	l	198	æ
7	39	'	71	G	103	ü	135	â	167	l	199	æ
8	40	(	72	H	104	ü	136	á	168	l	200	æ
9	41	)	73	I	105	ü	137	â	169	l	201	æ
10	42	*	74	J	106	ü	138	á	170	l	202	æ
11	43	+	75	K	107	ü	139	â	171	l	203	æ
12	44	,	76	L	108	ü	140	á	172	l	204	æ
13	45	-	77	M	109	ü	141	â	173	l	205	æ
14	46	.	78	N	110	ü	142	á	174	l	206	æ
15	47	/	79	O	111	ü	143	â	175	l	207	æ
16	48	0	80	P	112	ü	144	á	176	l	208	æ
17	49	1	81	Q	113	ü	145	â	177	l	209	æ
18	50	2	82	R	114	ü	146	á	178	l	210	æ
19	51	3	83	S	115	ü	147	â	179	l	211	æ
20	52	4	84	T	116	ü	148	á	180	l	212	æ
21	53	5	85	U	117	ü	149	â	181	l	213	æ
22	54	6	86	V	118	ü	150	á	182	l	214	æ
23	55	7	87	W	119	ü	151	â	183	l	215	æ
24	56	8	88	X	120	ü	152	á	184	l	216	æ
25	57	9	89	Y	121	ü	153	â	185	l	217	æ
26	58	:	90	Z	122	ü	154	á	186	l	218	æ
27	59	;	91	[	123	ü	155	â	187	l	219	æ
28	60	<	92	\	124	ü	156	á	188	l	220	æ
29	61	=	93	]	125	ü	157	â	189	l	221	æ
30	62	>	94	^	126	ü	158	á	190	l	222	æ
31	63	?	95	_	127	ü	159	â	191	l	223	æ

Hard Disk Drive Jumper Settings



TYPES OF INTERFACE

ESDI

- Enhance small device interface
- was a disc interface designed by [Maxtor Corporation](#) in the early 1980s
- dual cable 20/34 pin interface from controller to drive
- Data transfer rates is as high as 24 Mega-bits/sec
- Can be used with hardisk , floppy disk drives and tape drives to provide hih speed



- Pronounce as scuzzy
- was derived from "SASI", the "[Shugart Associates](#) System Interface", introduced by that company in 1981
- **Small Computer Systems Interface**
- are usually [ribbon cables](#) that have multiple 68 pin or 50 pin connectors
- Data Transfer Rate
Up to 160 MB/sec
- Standard high speed parallel interface
- Used for connecting microcomputer devices (hard disk, printers and LAN)

SCSI

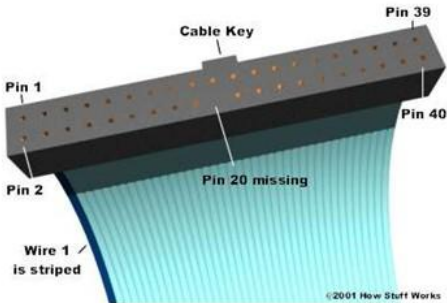
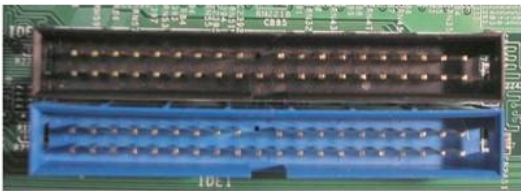


Parallel SCSI



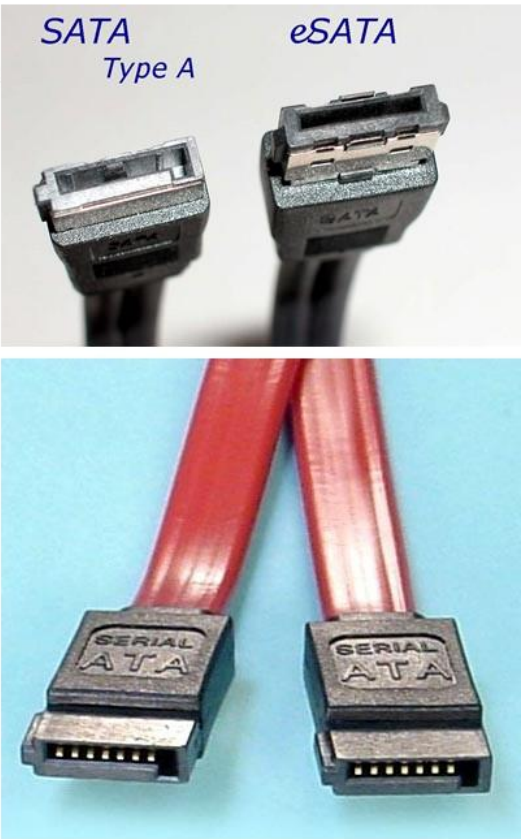
- **Advanced Technology Attachment**
- Despite the official name "ATA", this standard is better known by the commercial term IDE (*Integrated Drive Electronics*) or Enhanced IDE (EIDE or E-IDE)
- was developed on May 12, 1994 by the ANSI
- The ATA standard was originally intended for connecting hard drives
- ATAPI (ATA Packet Interface) was developed in order to be able to interface other storage peripherals ([CD-ROM drives](#), [DVD-ROM drives](#), etc.)
- Data Transfer Rate
Up to 133 MB/sec

ATA



SATA

- **Serial Advanced Technology Attachment**
- **primarily designed for transfer of data between a computer and mass storage devices such as hard disk drives and optical drives.**
- data transfer rates as high as 300 MB/s



ESATA

- External Serial Advanced Technology Attachment
- Before eSATA, external hard drives were connected via USB or FireWire. eSATA can provide data transfer rates significantly faster than USB or FireWire
- eSATA drives are hot-pluggable, meaning that you can attach or remove an eSATA hard drive from a system without powering down the system or the drive.

